



Handout 6.3: Waste Audit Data Sheet and Scale Up

Name: _____ Date: _____ Period: _____

Print pages 1 to 3 for every student. Page 4 is the sample data set for classes that could not collect their own, and for grade 6. Use the **grams** column if you weighed, and the **items** column if you counted. Every calculation works the same way in either unit.

Part 1. Where food waste happens

Where	One real example	About what share of all food waste
On the farm		
At the store		
At home		
In cafeterias and restaurants		

Circle the one that surprised you.

My guess before the lesson: _____ pounds per person per year. The real number is about: _____ pounds per person per year.

Part 2. Our audit

Our method (circle one): we weighed / we counted items

Days collected: _____ **Number of people the bins came from:** _____

Bin	Weight in grams	Weight in ounces	Number of items
EDIBLE (somebody could have eaten it)			
NOT EDIBLE (peels, cores, bones, grounds)			
TOTAL			

The edible bin is the one we do the math on. Peels and cores were never going to be eaten, so they are not the number we are trying to change.

The one thing in the edible bin that bothered me most: _____



Part 3. The math

Show your work in every box. Write the unit (grams, ounces, pounds, or items) after every answer.

Calculation 1: per person per day

Total edible waste _____ divided by number of people _____ = _____ per person for all the days

Then divide by the number of days _____ = _____ **per person per day**

Work:

Line 1: _____

Line 2: _____

Calculation 2: per person per school year

Per person per day _____ times 180 school days = _____ **per person per school year**

Work:

Line 1: _____

Convert to pounds: divide grams by 454, or divide ounces by 16.

_____ **pounds per person per school year**

Calculation 3: our class, then the whole school

Use the pounds number from Calculation 2.

Pounds per person per school year _____ times number of people in our class _____ = _____ **pounds per school year, our class**

Pounds per person per school year _____ times number of students in the building _____ = _____ **pounds of edible food per school year, whole school**

Work:

Line 1: _____

Line 2: _____

Check your units. Circle the unit of your final answer: grams / ounces / pounds / items

Part 4. Three changes

Each change has to name **who** does **what** and **when**. “Waste less” is not a change.

#	Who	What they do	When
1			
2			
3			

Which of your three costs nothing? ____

Which one would still be happening a month from now? ____ Why that one?

Answer: _____

Class vote result, the three changes we are actually doing:

1. _____
2. _____
3. _____

Exit card

Our class wastes about _____ pounds a year, and the one change I will actually do is _____.

Page 4: Sample data set

Use this if your class could not collect its own data. Say out loud that it is sample data and not ours.

A class of 24 students collected food scraps for 2 days. EDIBLE bin: 1,440 grams total. NOT EDIBLE bin: 2,160 grams total. The building has 600 students.

Work it through:

1,440 divided by 24 = ____ grams per person for 2 days ____ divided by 2 = ____ grams per person per day ____ times 180 = ____ grams per person per school year ____ divided by 454 = about ____ pounds per person per school year ____ times 24 students = about ____ pounds for the class ____ times 600 students = about ____ pounds for the whole school

Grade 6 version: do the first four lines only. Stop before the class line.

Page 5: Grade 8 stretch, put a dollar on it

Use an average food price per pound from this week's store circular.

Average price per pound I am using: \$_____ (what food? _____)

Question	Work	Answer
What does one person's yearly edible waste cost?		\$
What does the whole school's yearly edible waste cost?		\$
What could the FACS room buy with that?		

One sentence: the number that surprised me most was _____ because _____.